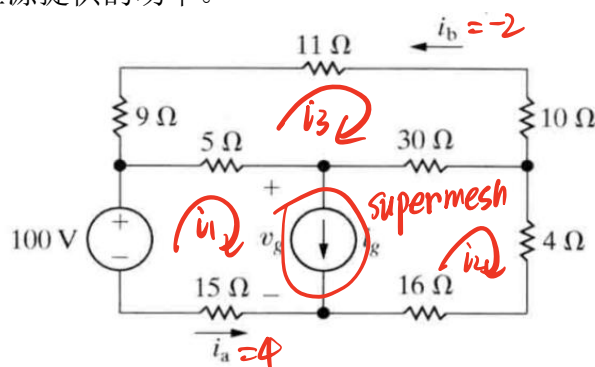


一、下图所示电路中的 $i_a = 4\text{ A}$, $i_b = -2\text{ A}$, 求:

- (1) 电流源的输出电流 i_g ;
- (2) $30\ \Omega$ 电阻上消耗的功率;
- (3) 100 V 电压源提供的功率。



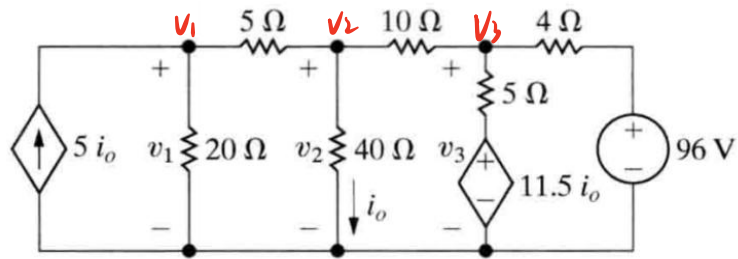
$$(1) \quad \begin{cases} -100 + 5(i_1 - i_3) + 30(i_2 - i_3) + 20i_2 + 15i_1 = 0 \\ 30i_3 + 30(i_3 - i_2) + 5(i_3 - i_1) = 0 \\ i_1 = -i_a = -4 \\ i_3 = -i_b = 2 \end{cases} \quad \begin{aligned} &\Rightarrow i_2 = 5\text{ A} \\ &i_g = i_1 - i_2 = -4 - 5 = -9\text{ A} \end{aligned}$$

$$(2) \quad P = R(i_2 - i_3)^2 = 30 \times (5 - 2)^2 = 270\text{ W}$$

$$(3) \quad P = U i_1 = 100 \times (-4) = -400\text{ W}$$



二、求下图所示电路的节点电压 v_1 , v_2 , v_3 。

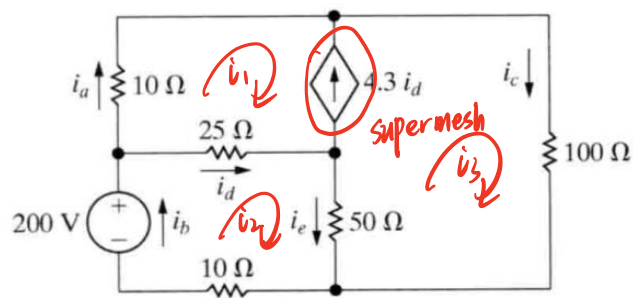


$$\begin{cases} -5i_o + \frac{v_1 - 0}{20} + \frac{v_1 - v_2}{5} = 0 \\ \frac{v_2 - v_1}{5} + \frac{v_2 - 0}{40} + \frac{v_2 - v_3}{10} = 0 \\ \frac{v_3 - v_2}{10} + \frac{v_3 - 11.5i_o}{5} + \frac{v_3 - 96}{4} = 0 \\ v_2 = 40i_o \end{cases}$$

$$\Rightarrow \begin{cases} (\frac{1}{20} + \frac{1}{5})v_1 + (-\frac{1}{5})v_2 + 0 = 0 \\ (-\frac{1}{5})v_1 + (\frac{1}{40} + \frac{1}{10})v_2 + (-\frac{1}{10})v_3 = 0 \\ 0 + (-\frac{1}{10} - \frac{11.5}{200})v_2 + (\frac{1}{10} + \frac{1}{5} + \frac{1}{4})v_3 = 24 \end{cases}$$

$$\Rightarrow v_1 = 156 \quad v_2 = 120 \quad v_3 = 78$$

三、用网孔电流法求下图所示电路中的支路电流 $i_a \sim i_e$ 。

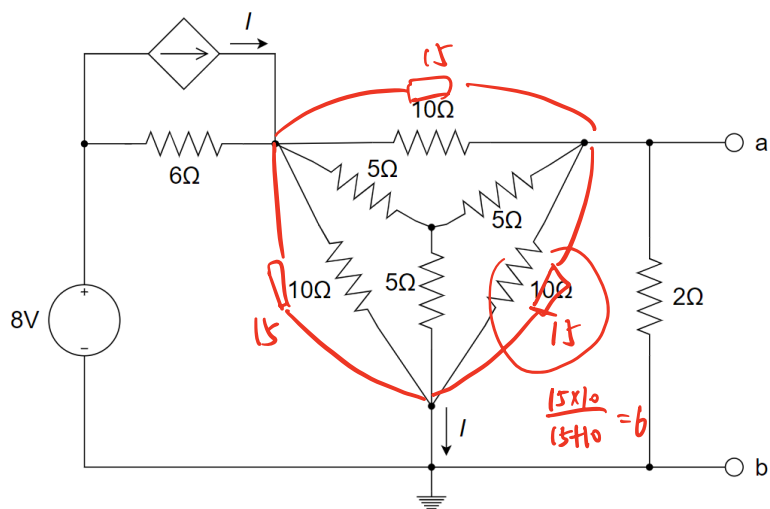


$$\Rightarrow \begin{cases} i_a = 5.7 \\ i_b = 4.6 \\ i_c = 0.97 \\ i_d = 4.6 - 5.7 = -1.1 \\ i_e = 4.6 - 0.97 = 3.63 \end{cases}$$

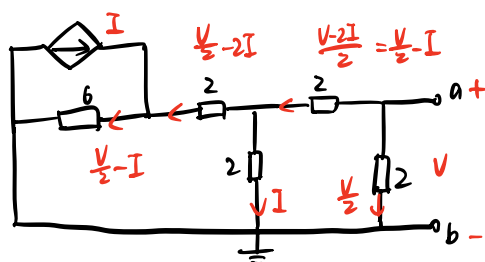
$$\begin{cases} 10i_1 + 100i_3 + 50(i_3 - i_2) + 25(i_1 - i_2) = 0 \\ -200 + 25(i_2 - i_1) + 50(i_2 - i_3) + 10i_2 = 0 \\ i_3 - i_1 = 4.3(i_2 - i_1) \end{cases} \Rightarrow \begin{cases} 35i_1 - 75i_2 + 150i_3 = 0 \\ -25i_1 + 85i_2 - 50i_3 = 200 \\ 3.3i_1 - 4.3i_2 + i_3 = 0 \end{cases}$$

$$\Rightarrow i_1 = 5.7A \quad i_2 = 4.6A \quad i_3 = 0.97A$$

四、计算并画出下图所示电路的戴维南等效电路。



明确所设变量含义
不要直接看形式套公式



$$6\left(\frac{V}{2} - I\right) + 2\left(\frac{V}{2} - 2I\right) - 2I = 0$$

$$\Rightarrow 3V - 6I + V - 4I - 2I = 0$$

$$\Rightarrow 4V = 12I \Rightarrow V = 3I$$

$$R_{Th} = \frac{V}{I} = 3\Omega \quad \times$$

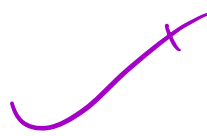
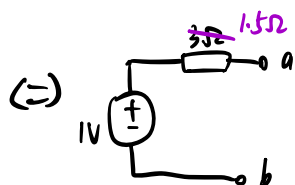
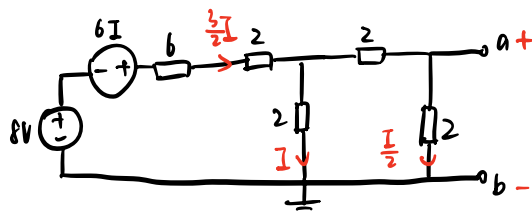
$$R_{Th} = \frac{V}{\frac{V}{2} - I + \frac{V}{2}} = 1.5\Omega$$

$$-(8 + 6I) + \frac{3}{2}I \times 8 + 2I = 0$$

$$\Rightarrow -4I - 8 + 12I = 0$$

$$\Rightarrow 8 = 8I \quad I = 1A$$

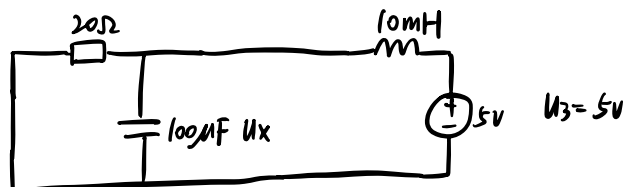
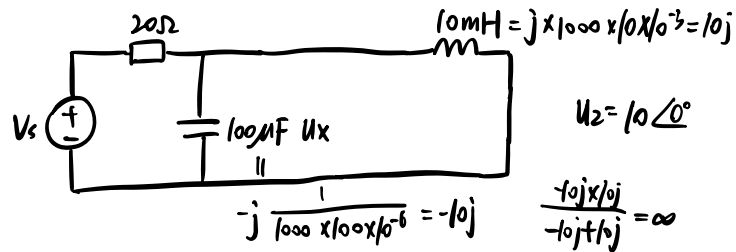
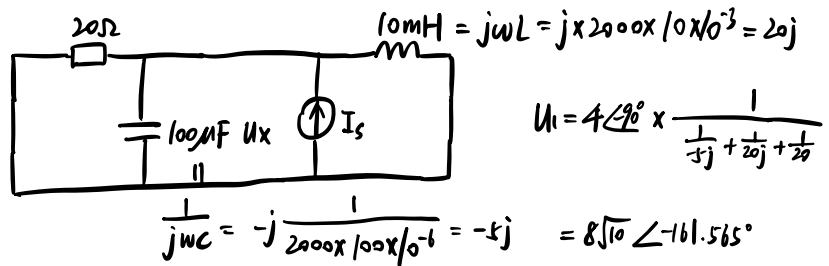
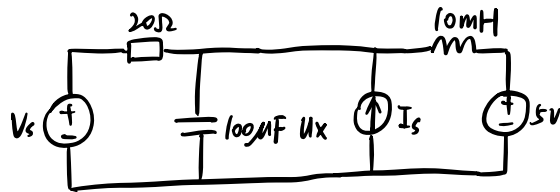
$$U_{Th} = 2 \times \frac{I}{2} = 1V$$



$$= 10 \angle 0^\circ$$

$$= 4 \angle -9^\circ$$

五、如下图所示电路，已知 $v_s = 10 \cos(1000t) \text{ V}$ ， $i_s = 4 \sin(2000t) \text{ A}$ ，用叠加原理求电容两端的电压 $u_x(t)$ 。

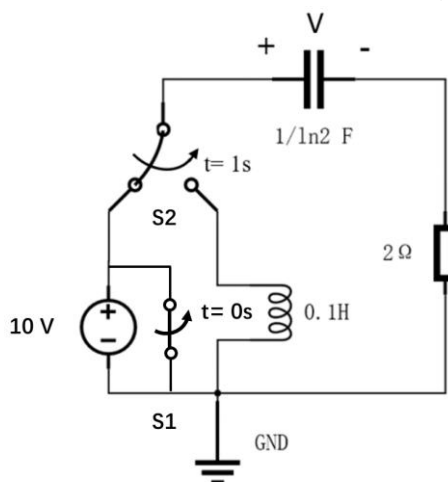


$$u_x = U_1 + U_2 + U_3 = 25.298 \cos(2000t - 161.565^\circ) + 10 \cos(1000t) + 5$$

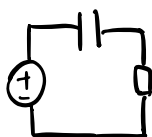
$$\sin(2000t - 71.565^\circ)$$

$$-9^\circ + 9^\circ$$

六、如下图所示电路，电容初始储能为 0，在 $t=0s$ 时开关 S1 开路， $t=1s$ 时刻，开关 S2 转接到另一端，求 $0 < t \leq 1s$ 和 $t > 1s$ 时电容两端电压 V 。



① $0 < t \leq 1$



$1 < 5\tau$ 未充

$$\tau = RC = 2 \times \frac{1}{\ln 2}$$

$$V_C(t) = 10 + (0 - 10)e^{-\frac{t}{\tau} \ln 2} = 10(1 - e^{-\frac{\ln 2}{2}t})$$

$$V_S = 10 \quad V_0 = 0$$

② $t > 1$



$$\alpha = \frac{R}{2L} = \frac{2}{2 \times 0.1} = 10 > \omega = \sqrt{\frac{1}{LC}} = \sqrt{\frac{\ln 2}{0.1}} \approx 2.6$$

过阻尼

$$V_0 = V_{C(1)} = 10(1 - e^{-\frac{\ln 2}{2}}) = 2.93 \quad V_S = 0$$

$$V_C(t) = V_S + A_1 e^{s_1(t-1)} + A_2 e^{s_2(t-1)}$$

$$\begin{cases} A_1 + A_2 = V_0 \\ A_1 s_1 + A_2 s_2 = 0 \end{cases}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega^2} = -0.351$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega^2} = -17.656$$

$$A_1 = 2.983$$

$$A_2 = -\frac{2.93}{5300} = -0.053$$

$$V_C(t) = 2.983 e^{-0.351(t-1)} - 0.053 e^{-17.656(t-1)}$$

$$V(t) = \begin{cases} 10(1 - e^{-\frac{\ln 2}{2}t}) & 0 < t \leq 1 \\ 2.983 e^{-0.351(t-1)} - 0.053 e^{-17.656(t-1)} & t > 1 \end{cases}$$

常用公式：

YΔ电阻关系	ΔY 电阻关系
$R_{12} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$	$R_1 = \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}}$
$R_{23} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$	$R_2 = \frac{R_{23} R_{12}}{R_{12} + R_{23} + R_{31}}$
$R_{31} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$	$R_3 = \frac{R_{31} R_{23}}{R_{12} + R_{23} + R_{31}}$

一阶电路全响应 $f(t) = f'(t) + [f(0_+) - f'(0_+)]e^{-\frac{t}{\tau}}$

二阶电路响应：

$$R > 2\sqrt{\frac{L}{C}} \quad v_C = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

$$R = 2\sqrt{\frac{L}{C}} \quad v_C = k_1 e^{-\alpha t} + k_2 t e^{-\alpha t}$$

$$R < 2\sqrt{\frac{L}{C}} \quad v_C = k e^{-\alpha t} \sin(\omega t + \beta)$$